

UI/UX Design Specification

Project: Autonomous Operations Dashboard

Design Philosophy: Warm Minimalism

Date: June 21, 2026

This specification outlines a front-end dashboard designed to be built quickly using a framework like Streamlit. The interface is engineered to present dense operational data—pricing thresholds, database logs, and generated text—without overwhelming the user.

The aesthetic focuses entirely on simplicity and sophistication. It avoids stark, clinical "developer tool" themes in favor of a warm, premium feel that aligns with a high-end retail environment.

1. Core Design Principles

- **Low Cognitive Load:** The user should immediately see what needs their attention (e.g., margin alerts, SKU errors) without hunting through complex menus.
- **Tactile Warmth:** Moving away from stark whites and harsh blues. We will use soft, warm undertones to make the interface feel inviting while maintaining a highly professional, high-end structure.
- **Typography over Ornamentation:** No heavy drop shadows, no gradients, and no unnecessary borders. The hierarchy is established entirely through font weight, spacing, and subtle color shifts.

2. Color Palette

The color system uses a warm, neutral foundation with muted accents to communicate system status without visual shouting. The inclusion of soft champagne and bronze tones subtly nods to two-tone material aesthetics while keeping the interface grounded.

Element	Color Name	Hex Code	Usage
Background	Warm Alabaster	#FAFAFA	Main application background. Reduces eye strain compared to pure white.
Surface/Cards	Cashmere White	#FFFFFF	For data cards and agent output containers.

Element	Color Name	Hex Code	Usage
Primary Text	Deep Charcoal	#2B2B2B	Headers, primary data readouts, and main copy. Softer than pure black.
Secondary Text	Warm Stone	#7A7A7A	Timestamps, minor labels, and secondary UI text.
Primary Accent	Muted Bronze	#B89B72	Primary action buttons (e.g., "Run Agents") and active tab underlines.
Success/Pass	Soft Sage	#8CA58A	Agent B logging a successful SKU validation.
Error/Alert	Dusty Rose	#C48787	Agent A flagging a margin drop; Agent B catching a miscategorized item.

3. Typography System

A combination of a refined serif for headers and a highly readable sans-serif for dense data creates a balanced, premium look.

- **Primary Headers:** *Playfair Display* (or similar elegant serif). Used only for the main dashboard title and primary section headers.
- **Body & Data:** *Inter* or *Geist*. Clean, monolinear, and highly legible at small sizes for reading tabular data, SKU numbers, and agent logs.

4. Layout Architecture & User Flow

The application will be a single-page, vertically scrolling dashboard.

A. The Global Header

- **Left:** Minimal text logo or title (e.g., "Operations Console").
- **Right:** A single, prominent primary button: "Initiate Audit Cycle".
- **Below:** A subtle timestamp showing the "Last synced" time.

B. Section 1: The Market Monitor (Agent A)

- **Visual Structure:** A horizontal row of KPI cards.

- **Content:** Card 1: Current Platinum Spot Price (Large typography, dynamic indicator arrow). Card 2: Margin Health.
- **Alert State:** If a margin drops below the threshold, a table appears below the cards listing the affected items, highlighted subtly in Dusty Rose.

C. Section 2: Catalog & SKU Auditor (Agent B)

- **Visual Structure:** A clean, borderless terminal-style log or a simple data table.
- **Content:** A chronological feed of the agent's actions.
- **Logic Display:** The UI must clearly differentiate standard approvals from structural corrections. For example, if the agent detects a two-tone piece grouped under a pure platinum SKU, the log entry should be flagged with an error icon, explicitly stating: *"Upload Blocked: Two-tone item identified. Must be generated as a distinct SKU."*

D. Section 3: Brand Voice Output (Agent C)

- **Visual Structure:** A masonry grid or a clean vertical list of "Draft Cards."
- **Content:** Displays the generated metadata and product descriptions for newly validated SKUs.
- **Interaction:** Each card features a "Copy to Clipboard" icon (minimalist outline style) so the user can easily pull the approved, sophisticated copy into the live storefront.

5. Component Styling (Streamlit Adaptation)

If building this in Streamlit, you will use custom CSS injection (`st.markdown("<style>...</style>", unsafe_allow_html=True)`) to override the default clinical look:

- Remove the default Streamlit hamburger menu and footer for a cleaner app feel.
- Apply subtle `border-radius: 8px` to all data cards (avoiding perfectly sharp corners to maintain warmth).
- Add a subtle padding (`padding: 24px`) around text blocks so the generated copywriting has room to breathe.